Lahore Garrison University

Department of Criminology

Final Year Project Proposal

Student Information:

Name:	**Email:**
Contact Information:	**Supervisor:**

Project Title:

Detection and Attribution of AI-Generated Synthetic Online Personas

1. Introduction

The rapid advancement of Artificial Intelligence has enabled the creation of highly realistic synthetic online personas across social media platforms, forums, and digital communities. These AI-generated identities are increasingly used in misinformation campaigns, social engineering attacks, influence operations, financial fraud, and coordinated cyber threats.

Traditional cybersecurity and digital forensic techniques primarily focus on malware detection and network intrusion; however, modern threats increasingly originate from **human-like online identities rather than malicious software**.

This project proposes an intelligent OSINT-based system capable of detecting and attributing AI-generated synthetic personas through behavioral analytics, stylometric analysis, and machine learning techniques.

The system aims to assist digital forensic investigators and cybersecurity analysts in identifying coordinated fake identities operating across online platforms

2. Problem Statement

Existing OSINT investigation methods rely heavily on manual analysis, making it difficult to detect sophisticated AI-generated personas that mimic human communication patterns.

Current challenges include:

- AI accounts producing human-like text content
- Coordinated fake identity networks
- Cross-platform persona reuse
- Lack of automated attribution mechanisms
- Difficulty distinguishing automated influence campaigns from legitimate users

There is a need for an automated forensic framework capable of detecting synthetic identities using behavioral and linguistic evidence.

3. Project Objectives

The main objectives of this project are:

- Collect publicly available OSINT data from online platforms
- Extract behavioral and stylometric features from user activities
- Detect AI-generated synthetic personas using machine learning
- Attribute related fake accounts through clustering technique
- Provide forensic visualization and investigation report
- Assist cybersecurity analysts in threat intelligence operations

4. Scope of the Project

Included Scope

- Public OSINT data collection
- Textual behavior analysis
- Activity pattern analysis
- Machine learning classification
- Persona clustering & attribution
- Investigation dashboard visualization

Excluded Scope

- Private or unauthorized data access
- Platform exploitation
- Real-time national surveillance systems

The system will operate strictly using **ethical OSINT principles**.

5. Proposed Methodology

The system follows a multi-layer forensic intelligence pipeline.

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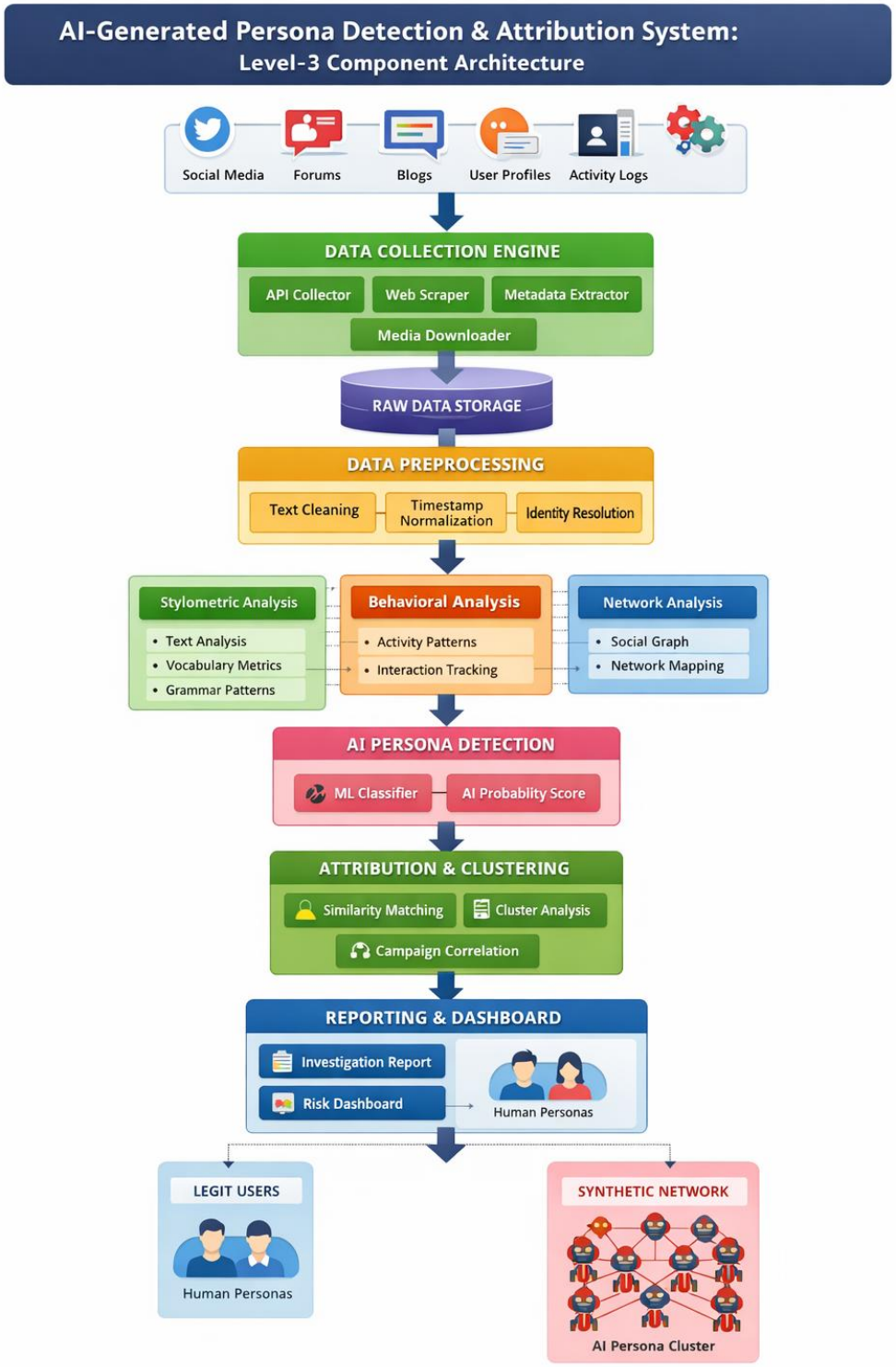

➢ Phase 1 — OSINT Data Acquisition

Publicly available data such as posts, comments, timestamps, and profile metadata will be collected using APIs and web scraping tools.

➢ Phase 2 — Data Preprocessing

Collected data will undergo:

- Text cleaning
- Noise removal
- Timestamp normalization
- Identity resolution
- Dataset structuring

➢ Phase 3 — Feature Extraction

◆ Stylometric Features

- Writing style patterns
- Vocabulary richness
- Sentence structure
- Grammar consistency

◆ Behavioral Features

- Posting frequency
- Activity timing
- Interaction behavior
- Automation patterns

◆ Network Features

- Interaction graphs
- Community relations
- Coordination detection

➢ Phase 4 — AI Persona Detection

Machine learning models will classify accounts as:

Human Persona
or
Synthetic Persona

Algorithms considered:

- Random Forest
- Support Vector Machine
- Isolation Forest

➢ Phase 5 — Attribution & Clustering

Detected synthetic accounts will be grouped using similarity analysis to identify coordinated campaigns or single operators controlling multiple identities.

➢ Phase 6 — Visualization & Reporting

A dashboard interface will present:

- Risk scores
- Persona clusters
- Behavioral evidence
- Investigation summaries

6. Expected Outcomes

The system will:

- Automatically detect AI-generated online identities
- Identify coordinated fake persona networks
- Provide digital forensic evidence support
- Assist cyber threat intelligence investigations
- Demonstrate practical OSINT automation capability

7. Significance of the Project

This project contributes directly to:

◆ Digital Forensics

- Identity attribution analysis
- Behavioral evidence generation
- Online investigation automation

◆ Cybersecurity

- Detection of influence operations
- Botnet-like social campaigns
- Social engineering prevention

◆ OSINT Intelligence

- Automated intelligence extraction
- Cross-platform persona correlation

9. Hardware Requirements

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Component	Minimum Requirement
Processor	Intel i5 / Ryzen 5
RAM	8 GB (16 GB Recommended)
Storage	256 GB SSD
GPU	Optional (for ML acceleration)
Internet	Stable broadband connection
System Type	64-bit machine

Software Requirements

8. Software Requirements

Category	Tools / Technologies
Programming Language	Python
Data Collection	SNScrape, Requests, BeautifulSoup
Data Processing	Pandas, NumPy
NLP Processing	NLTK, SpaCy
Machine Learning	Scikit-learn
Network Analysis	NetworkX
Visualization	Matplotlib, Plotly
Dashboard	Streamlit
Database	MongoDB / PostgreSQL
Development Environment	VS Code / PyCharm
Version Control	Git & GitHub
Operating System	Windows / Linux

10. Conclusion

The proposed system introduces a practical OSINT-driven framework for detecting and attributing AI-generated synthetic online personas. By combining behavioral analytics, stylometry, and machine learning, the project bridges the gap between digital forensics and modern cybersecurity threats emerging from AI-driven online ecosystems.

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Date of Submission: 28-2-2026	**Signature:** [Your signature]

Supervisor Remarks:	[Accepted] / [Rejected]